

ZeroDrift — What Is This Project? (the 3-minute read for anyone)

The problem, like a story

Internet through light. Instead of radio, two stations fire an invisible laser at each other — crazy fast, un-jammable, no spectrum licence. One catch: a laser beam is a thin thread of light. Both stations must point **exactly** at each other — with hair-thin accuracy — while wind shakes one and the other flies overhead at 27,000 km/h. If pointing slips, the link dies. Pointing is the hardest part, and ISRO (problem SIH26169) asked teams to build this "pointing brain" **in software**, without lakhs of rupees of optical hardware.

What is the "beacon"?

Every laser terminal carries a small helper light — the **beacon** — that blinks at an agreed speed (ours: 4 blinks per second). It is a lighthouse saying "I am here, I am the right one." Stars and sun-glints can be brighter, but they can never blink correctly — that blink is the password.

What did we build?

Two things, both finished and working: 1. **A virtual test-range** — a physics-true artificial sky: stars, clouds, atmospheric shaking, fake decoy lights, and a beacon moving on a *real* ISS orbit. Like a flight simulator, but for laser pointing. 2. **The pointing brain** — software that looks at that noisy camera picture, finds the beacon, checks its blink password, has a small self-trained AI double-check the lock, predicts where the beacon will be 40 milliseconds ahead, and steers a virtual camera to keep it centred — 30 times every second.

The brain never sees the answer. The simulator secretly knows where the beacon truly is and grades the brain afterwards — like an examiner with an answer key the student never touches. That is why our numbers are honest.

What can it do? (measured, not promised)

- Finds and locks the beacon in **1.27 seconds** from cold.
- Holds lock **98.3%** of an ISS pass; beacon in view **100%** of the time.
- Fooled by a wrong light **0 times in 64 random test runs**.
- Uses **~20 of every 33 milliseconds** on a normal laptop — no GPU, no cloud.
- Bottom line: link usable **99.3%** of the time with our brain, vs **14.7%** without.

Where can I see it?

- **Website** (any phone): zerodrift-fsoc-pat.netlify.app — replays a recorded run of the real engine, every number genuine.
- **Desktop app** — the live engine; judges can turn up turbulence and watch it fight and recover.
- **Code** — github.com/pranshukesavmishra/SIH-2026, open source, 73/73 tests passing.

Little dictionary

- **FSOC** — Free-Space Optical Communication: internet by laser through air/space.
- **μrad (micro-radian)** — a tiny tilt: ~ 1 mm sideways per km of distance.
- **Coarse alignment** — our job: get the beacon into view and hold it steady so the fast fine-steering mirror can do the last tiny bit.
- **COAST** — when the beacon blinks off, the brain keeps pointing from prediction. Not a failure — the plan.

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